### 5.3.14 I/O port characteristics

#### General input/output characteristics

Unless otherwise specified, the parameters given in Table 46 are derived from tests performed under the conditions summarized in Table 16. All I/Os are designed as CMOS and TTL-compliant.

*Note: For information on GPIO configuration, refer to the application note STM32 GPIO configuration for hardware settings and low-power consumption (AN4899).*

**Table 46. I/O static characteristics**

The I/O structure options listed in this table can be a concatenation of options including the option explicitly listed. For instance, TT_a refers to any TT I/O with _a option. TT_xx refers to any TT I/O and FT_xx refers to any FT I/O.
All I/Os are CMOS- and TTL-compliant (no software configuration required). Their characteristics cover more than the strict CMOS-technology or TTL parameters. The coverage of these requirements is shown in the figure below.

| Symbol | Parameter | Condition | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| $V_{IL}$ | I/O input low level voltage | 2.7 V<$V_{DDIOx}$<3.6 V | - | - | $0.3V_{DD}$ (1) | V |
| | I/O input low level voltage | | - | - | $0.4 V_{DD}$ - 0.1(2) | |
| $V_{IH}$ | I/O input high level voltage | 2.7 V < $V_{DDIOx}$ < 3.6 V | $0.7V_{DD}$ (1) | - | - | V |
| | I/O input high level voltage | | $0.52V_{DD}$ + 0.18(2) | - | - | |
| $I_{leak}$ (3) | FT_xx Input leakage current(2) | 0 < $V_{IN}$ ≤ Max($V_{DDXXX}$)(4) | - | - | ±-200 | nA |
| | | Max($V_{DDXXX}$) < $V_{IN}$ ≤ Max($V_{DDXXX}$ + 1 V)(5)(6)(4) | - | - | ±2500 | |
| | | Max($V_{DDXXX}$ + 1) < $V_{IN}$ ≤ 5.5 V (5)(6)(4) | - | - | 750 | |
| | TT_xx Input leakage current | 0 < $V_{IN}$ ≤ Max($V_{DDXXX}$ + 1 V)(4) | - | - | ±200 | |
| $R_{PU}$ | Weak pull-up equivalent resistor(7) | $V_{IN}$ = $V_{SS}$ | 30 | 40 | 50 | kΩ |
| $R_{PD}$ | Weak pull-down equivalent resistor(7) | $V_{IN}$ = $V_{DD}$ | 30 | 40 | 50 | |
| $C_{IO}$ | I/O pin capacitance | - | - | 5 | - | pF |

1. *Compliant with CMOS requirements.*
2. *Specified by design - Not tested in production.*
3. *This parameter represents the pad leakage of the I/O itself. The total product pad leakage is provided by the following formula: ITotal_Ileak_max = 10 µA + [number of I/Os where $V_{IN}$ is applied on the pad] × Ilkg max.*
4. *Max($V_{DDXXX}$) is the maximum value of all the I/O supplies.*
5. *To sustain a voltage higher than the minimum of $V_{DD}$ or $V_{DDA}$ plus 0.3 V, disable the internal pull-up and pull-down resistors.*
6. *$V_{IN}$ must be less than Max($V_{DDXXX}$) + 3.6 V.*
7. *The pull-up and pull-down resistors are designed with a true resistance in series with a switchable PMOS/NMOS. This PMOS/NMOS contribution to the series resistance is minimal (~10% order).*

All I/Os are CMOS- and TTL-compliant (no software configuration required). Their characteristics cover more than the strict CMOS-technology or TTL parameters. The coverage of these requirements is shown in the following figure.